METACHRONAL CASCADE FLAPPER

Eight-Wing Staggered Cascade Ornithopter

Drone Prototype — Engineering Build Proposal

Version 1.0 | Design Stage Document | 2025

Architecture	8 wings, 4 per side, staggered cascade
Wing profile	Selig S1223 — high-lift low-Reynolds
Flap frequency	1 Hz — power-optimal amplitude
Stroke amplitude	120 degrees full arc
Root degrees of freedom	3 per wing: flap, pitch, sweep
Wave strategy	Metachronal — 90 degree phase spacing
Enhancements	VGs + covert flaps + end plates + fences
Controller	Raspberry Pi 5 + dual PCA9685 + BNO085 IMU
Power	6S LiPo 22.2V, 5000–6000 mAh
Estimated MTOW	2.8 – 4.2 kg
Estimated endurance	18 – 32 minutes hover

This document presents a complete engineering build specification for the drone-scale prototype of the Metachronal Cascade Flapper — an eight-wing ornithopter using a staggered tandem cascade architecture with independent three-degree-of-freedom wing roots, metachronal wave propulsion, and a heterogeneous wing geometry optimised for each cascade position. All design decisions, component selections, build methods, and firmware architecture described herein are based on the accumulated analysis conducted throughout the design development process.

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CHAPTER 1

DESIGN OVERVIEW AND REQUIREMENTS

1.1 Project Objective

The objective of this build is to produce a flying prototype of the Metachronal Cascade Flapper architecture at drone scale. The prototype shall demonstrate sustained hover, controlled forward flight, reverse flight, and lateral translation using an eight-wing staggered tandem cascade driven by a metachronal wave controller. All eight wings shall operate independently through three-degree-of-freedom root mechanisms under software control.

1.2 Design Requirements

Maximum takeoff weight	4.2 kg maximum
Minimum hover endurance	15 minutes on single charge
Flap frequency	1.0 Hz nominal, 0.8–1.2 Hz variable
Stroke amplitude	120 degrees (plus or minus 60 degrees from centre)
Wing count	8 wings total, 4 per side
Wing root DoF	3 per wing: flap, pitch, sweep
Sweep range	30 degrees forward to 120 degrees aft (stow)
Pitch range	Minus 5 to plus 50 degrees angle of attack
Control interface	5-channel RC with FPV capability
Stability	IMU-stabilised on all three axes
Fault tolerance	Continued controlled flight with 6 of 8 wings active
Noise	Dominant frequency below 20 Hz — effectively inaudible

1.3 Key Design Decisions

The following fundamental decisions were reached through the aerodynamic analysis process and are treated as fixed constraints for this build:

• Selig S1223 airfoil chosen as optimal for Re 50,000–150,000 flapping flight

• Staggered root arrangement: each successive root positioned one chord length lower and further aft than the one preceding it

• Heterogeneous wing geometry: W1 through W4 are not identical — each is optimised for its aerodynamic role within the cascade

• 120 degrees stroke amplitude confirmed as the mathematically power-optimal amplitude for hovering flapping flight

• Series hybrid electronics architecture: central controller outputs kinematic waveforms to 24 independent servo channels

• Metachronal wave as primary lift strategy: 90-degree phase spacing between adjacent wings, giving near-continuous lift and minimal body oscillation

• Forward sweep during power stroke, aft sweep during recovery — directly replicating the kinematic strategy observed across all slow-flying vertebrates

CHAPTER 2

AERODYNAMIC ARCHITECTURE

2.1 The Cascade Principle

The eight wings are arranged in a staggered tandem cascade on each side of the fuselage. Each wing root is positioned approximately one chord length lower and one chord length further aft than the wing root immediately forward of it. This creates a diagonal staircase arrangement in side profile. The consequence is that each successive wing's stroke arc occupies a different spatial volume from its predecessor, dramatically reducing destructive aerodynamic interference compared to a flat co-planar arrangement.

The tandem interaction between wings is managed through two independent mechanisms: temporal separation via the metachronal phase wave, and spatial separation via the staggered root geometry. Together these constitute double phase separation — the most effective known strategy for operating multiple tandem flapping surfaces efficiently.

2.2 The Metachronal Wave

Rather than all wings flapping in unison, the controller imposes a traveling phase wave along the four-wing stack on each side. The default spacing is 90 degrees between adjacent wings, giving the following timing across one full cycle:

Wing	Phase Offset	Timing
Wing 1 (front)	Phase 0 degrees	Reference — starts power stroke first
Wing 2	Phase 90 degrees	Power stroke begins 250 ms after W1
Wing 3	Phase 180 degrees	Power stroke begins 500 ms after W1
Wing 4 (rear)	Phase 270 degrees	Power stroke begins 750 ms after W1

At any instant, one wing on each side is in mid power stroke, one is near reversal, one is in mid recovery, and one is approaching the next power stroke. Lift generation is therefore nearly continuous, with minimal oscillation transmitted to the fuselage.

2.3 Dynamic Sweep Schedule

Each wing's sweep angle varies continuously within each flap cycle. During the downstroke (power stroke) the wing sweeps progressively forward, reaching maximum forward sweep at mid-stroke. During the upstroke (recovery) the wing sweeps progressively aft, reaching near-stow position at mid-recovery. This pattern is consistent across all observed slow-flying vertebrate species and provides three distinct aerodynamic benefits:

- Maximum effective wingspan during the power stroke — highest CL, most lift
- Reduced effective area during recovery stroke — minimum drag, lower power consumption
- Inboard-directed spanwise flow during forward sweep — delays tip stall at high AoA

2.4 Wake Mode Library

Mode	Phase	Wave Type	Description
HOVER	180 degrees	Counter-phase pairs	Minimum body oscillation, efficient station-keeping
CRUISE	135 degrees	Near-metachronal	Low drag, smooth forward translation
CLIMB	90 degrees	Metachronal default	Best lift-efficiency for altitude gain
TAKEOFF	0 degrees	All in-phase burst	Maximum instantaneous lift — brief duration
REVERSE	270 degrees	Wave direction flip	Thrust vectored rearward — software only
DESCEND	180 degrees	Symmetric drag AoA	Controlled sink rate
SPEED	120 degrees	Wings partly swept	Reduced drag, higher advance ratio

CHAPTER 3

WING GEOMETRY SPECIFICATION — ALL EIGHT WINGS

The eight wings are not identical. Each wing is optimised for the specific aerodynamic environment it encounters within the cascade. The reference span S is defined as the span of Wing 1. All other wing dimensions are stated as multiples of this reference. For a 3.5 kg build target, S is set at 550 mm.

Wing 1 — Cascade Leader (Frontmost)

Role	Primary lift generation, LEV initiation, organised wake for cascade
Aspect ratio	4.0
Span (per side)	550 mm (reference span S)
Root chord	138 mm
Tip chord	76 mm (elliptical taper)
Taper ratio	1.8 : 1
Planform	Elliptical taper — maximum chord at 40% semi-span
Dihedral	+3 degrees
Forward sweep (max)	30 degrees forward at mid-stroke
Aft sweep (stow)	110 degrees aft — wing alongside fuselage
Wing fence	One fence at 45% semi-span
End plate	Symmetric NACA 0008, 12% semi-span height above and below
Flexibility	Low — stiff carbon spar, semi-rigid ribs

Wing 2 — Wake Conditioner

Role	Capture W1 trailing edge vortex, condition flow for W3

Aspect ratio	3.5
Span (per side)	495 mm (0.90 x S)
Root chord	141 mm (wider than W1 — intercepts more wake)
Tip chord	82 mm
Taper ratio	1.7 : 1
Planform	Near-rectangular with rounded tips
Dihedral	+5 degrees
Root offset (below W1)	Approx 138 mm — one W1 chord length lower
Root offset (aft of W1)	Approx 138 mm — one W1 chord length aft
Wing fence	One fence at 40% semi-span
End plate	90% of W1 end plate area
Flexibility	Moderate — allow leading edge passive flex for wake capture

Wing 3 — Primary Thrust Generator

Role	Exploit upwash from outer W1 vortex, generate primary forward thrust
Aspect ratio	5.0 (highest in stack)
Span (per side)	578 mm (1.05 x S — longest wing)
Root chord	116 mm (narrower — high AR)
Tip chord	58 mm
Taper ratio	2.0 : 1
Planform	Progressive taper with slight aft sweep at tip
Dihedral	+7 degrees (highest in stack)
Root offset (below W2)	Approx 138 mm lower and aft
Wing fences	Two fences at 30% and 60% semi-span
End plate	110% of W1 area — largest in stack
Flexibility	Low to moderate — matched structural stiffness

Wing 4 — Cascade Terminus (Rearmost)

Role	Wake energy recovery, cascade termination, minimum drag on recovery
Aspect ratio	4.5
Span (per side)	523 mm (0.95 x S)
Root chord	116 mm (wide root for wake capture)
Tip chord	46 mm
Taper ratio	2.5 : 1 (most strongly tapered — widest root in stack)
Planform	Strong taper — wide at root, narrow at tip
Dihedral	+4 degrees
Root offset (below W3)	Approx 138 mm lower and aft
Wing fence	One fence at 50% semi-span
End plate	Standard — same area as W1
Flexibility	Highest in stack — passive aeroelastic adaptation to complex inflow

NOTE: Wings on the right side are mirror images of the left side. Root position offsets accumulate: W4 root is positioned approximately 414 mm below and 414 mm aft of the W1 root on each side, forming the diagonal cascade staircase.

CHAPTER 4

AIRFOIL, ENHANCEMENTS, AND SURFACE DETAILS

4.1 Airfoil Selection: Selig S1223

The Selig S1223 is selected as the wing section for all eight wings of the drone prototype. It was designed at the University of Illinois Urbana-Champaign using a concave pressure recovery philosophy with aft loading, specifically targeting Reynolds numbers between 50,000 and 500,000. Its principal characteristics make it uniquely suitable for this application:

Maximum lift coefficient	2.36 at Re 450,000 — highest of tested comparable airfoils
Thickness	12.1% chord
Maximum camber	8.7% chord at 47% chord position
Stall character	Gradual, progressive — no abrupt separation
High AoA tolerance	Maintains useful lift well beyond conventional stall angles
Bird wing comparison	Closely matches inner wing geometry of seagull and merganser
Ornithopter validation	Independently validated for flapping wing flight applications
VG test data	Tested with VGs at 17% chord by original designer (Selig, 1997)

4.2 Spanwise Airfoil Variation

The S1223 section is not applied uniformly from root to tip. A spanwise blend reduces camber and thickness progressively toward the tip:

Root to 30% semi-span	S1223 full section: 12.1% thickness, 8.7% camber
30% to 70% semi-span	Blended: thickness 11.5%, camber 7.5% — reduced pitching moment

70% to 100% semi-span	Tip section: 10.0% thickness, 6.5% camber — minimum tip mass

4.3 Vortex Generators

Counter-rotating vane pairs are mounted on the upper surface of each wing. The array extends from 20% to 85% semi-span, leaving the root zone clear of interference with the root hinge mechanism and the tip zone clear for end plate integration.

Type	Counter-rotating vane pairs
Chord position	17% chord on W1 (per Selig 1997 test); adjusted per wing — see below
VG height	0.8% chord — matched to local boundary layer thickness
Vane cant angle	18 degrees to local free stream
Pair spacing	4 x VG height between vanes in a pair
Array spacing	8 x VG height between pair centres
Span coverage	20% to 85% semi-span on each wing
Material	0.3 mm carbon fibre sheet, bonded with structural film adhesive

Wing	VG Position	Rationale
W1	17% chord	Clean inflow — validated placement
W2	15% chord	Moved forward: W1 downwash reduces local AoA at W2
W3	20% chord	Moved aft: higher local Re from W2 wake energisation
W4	22% chord	Furthest aft: most complex inflow, lowest local AoA

4.4 Covert Flaps

Three rows of passively-hinged covert-inspired flaps are mounted on the upper surface of each wing. These deploy automatically when the local angle of attack exceeds the separation threshold, extending the effective operating AoA range to 35–42 degrees. No actuation is required — deployment is entirely passive. The flaps lie flat during the feathered recovery stroke due to reduced local pressure.

Row	Chord Position	Deploys at	Mechanism and Wings
Row 1	35–40% chord	AoA > 15 degrees	Pressure dam effect — W1, W3, and W4 only
Row 2	50–55% chord	AoA > 20 degrees	Shear layer interaction — all wings
Row 3	65–70% chord	AoA > 28 degrees	Secondary pressure recovery — W1, W3, and W4 only

Flap chord (each row)	6–8% of local wing chord
Span coverage	15% to 85% semi-span per row
Material	0.3 mm carbon fibre sheet
Hinge	Piano wire, 0.5 mm diameter, full-span along row
Spring preload	Light leaf spring — calibrated deployment threshold per row
Retention	Magnetic catch limits flutter at speed — releases at AoA threshold

4.5 End Plates

Symmetric end plates are fitted at each wing tip. Each plate presents equal area above and below the wing chord plane to avoid generating asymmetric roll or dihedral moments. The plates retain tip vortex energy at the wing tip rather than shedding it into the downwash corridor of the next wing.

Section	NACA 0008 — symmetric, thin
Area (per plate)	15% of wing plan area for W1, W3; 12% for W2, W4
Plate height	10% of semi-span above and below chord plane
Chord length	Equal to local wing tip chord
Material	0.5 mm carbon fibre sheet, CNC cut
Attachment	Integrated with tip spar — bonded and bolted

4.6 Wing Fences

Spanwise wing fences prevent the chordwise boundary layer from migrating inboard during high angle of attack operation, which would cause premature tip stall. Each fence divides the wing span into independent aerodynamic zones, each developing its own leading edge vortex independently.

Height	3% of local chord
Chord extent	From 10% to 72% chord
Material	0.2 mm carbon fibre sheet, sharp leading edge
W1 position	One fence at 45% semi-span
W2 position	One fence at 40% semi-span
W3 positions	Two fences at 30% and 60% semi-span — longest wing, two zones
W4 position	One fence at 50% semi-span

CHAPTER 5

WING ROOT MECHANISMS

5.1 Mechanism Overview

Each of the eight wings is attached to the fuselage through a three-degree-of-freedom root mechanism. The mechanism must simultaneously provide the following independent motions, each driven by its own dedicated servo:

Axis	Range	Function
Degree of Freedom 1: Flap	Up-down oscillation, plus or minus 60 degrees	Primary lift and thrust generation — the stroke axis
Degree of Freedom 2: Pitch	Angle of attack control, minus 5 to plus 50 degrees	Feathers wing on recovery stroke, sets AoA on power stroke
Degree of Freedom 3: Sweep	Forward to aft sweep, 30 degrees fwd to 110 degrees aft	Performance optimisation and stow function — full aft is stowed

5.2 Mechanism Architecture

The three axes are arranged as nested gimbals. The sweep axis is outermost — a horizontal pivot on the fuselage attachment point. The flap axis is the middle gimbal — a horizontal pivot perpendicular to the fuselage centreline. The pitch axis is innermost — a pivot running spanwise along the wing root chord. This nesting ensures that all three axes are fully independent and do not mechanically interfere with each other across their full operating ranges.

5.3 Servo Selection

Each of the three axes requires a different servo specification reflecting the different loads and speed requirements of that axis:

Axis	Requirement	Recommended Servo	Rationale
Flap axis	High torque, moderate speed	KST X12-508 or equivalent: 8.0 kg.cm at 7.4V, 0.08 sec/60 deg, 22g	Handles peak aerodynamic and inertial loads at stroke reversal
Pitch axis	Moderate torque, HIGH speed	KST X08H or equivalent: 3.5 kg.cm at 7.4V, 0.05 sec/60 deg, 16g	Must complete 40-degree pitch flip in under 80 ms at stroke reversal
Sweep axis	Moderate torque, slow	Hitec HS-5125MG or equivalent: 3.5 kg.cm, 0.17 sec/60 deg, 24g	Sweep changes slowly within cycle — speed less critical than holding torque

Total servo count: 24 (8 wings x 3 axes). Total servo mass estimate: 480–520 g. All servos operate from the 6S LiPo via a 7.4V BEC (battery eliminator circuit). Servos are specified with metal gears and dual ball bearings as minimum standard given the cyclic load environment.

5.4 Mechanism Materials

Gimbal frame	3 mm carbon fibre plate, CNC cut
Pivot pins	3 mm diameter silver steel rod, hardened
Pivot bearings	3x7x3 mm miniature ball bearings, stainless
Servo horns	Aluminium, machined to suit geometry
Push rods (flap)	2 mm diameter carbon rod with M2 ball link ends
Push rods (pitch)	1.5 mm diameter carbon rod, M2 ball links
Wing attachment plate	2 mm carbon fibre, bonded to wing LE spar
Fasteners	M2 and M3 stainless steel socket head cap screws
Adhesive	Structural epoxy (Araldite 2011 or equivalent) for CFRP joints

CHAPTER 6

FUSELAGE AND STRUCTURAL DESIGN

6.1 Fuselage Layout

The fuselage is a structural spine onto which all wing root attachment points, electronics bay, battery bay, and payload bay are mounted. The staggered cascade geometry means the fuselage must provide eight root attachment points arranged in a diagonal staircase pattern on each side — four attachment points descending aft on the left and four on the right.

Overall length	620–720 mm
Width at widest point	80–100 mm (electronics bay)
Height at tallest point	90–120 mm (root attachment zone)
Root attachment span	W1 to W4 roots span 414 mm aft and 414 mm vertically
Electronics bay	Forward section, 80 x 60 x 50 mm
Battery bay	Central, removable hatch, accommodates 6S 5000 mAh pack
Payload bay	Aft section, 60 x 40 x 30 mm for FPV camera or sensors
Landing gear	Fixed skid type, carbon tube, 15 mm ground clearance

6.2 Structural Materials

Primary spine	25 x 10 mm carbon fibre rectangular tube, 2 mm wall
Root attachment frames	3 mm carbon fibre plate, CNC cut, bonded to spine
Cross-members	10 mm carbon square tube connecting left and right sides
Equipment tray	1.5 mm carbon fibre sheet

Fairing (optional)	1 mm fibreglass moulded shell over foam core — cosmetic only
Total structural mass (estimated)	180–240 g

6.3 Wing Construction

Each wing is constructed as a traditional spar-and-rib assembly with a flexible membrane covering. The construction sequence for each wing is as follows:

- Leading edge spar: 4 mm carbon fibre round tube, full span. This is the primary structural member and the attachment point for the root pitch axis.
- Secondary spar at 60% chord: 3 mm carbon rod, provides torsional stiffness.
- Ribs: 0.8 mm carbon fibre sheet, CNC cut to S1223 profile with spanwise blending. Rib pitch: 40 mm from root to 70% span; 55 mm from 70% to tip.
- Membrane: 20-micron ripstop nylon, heat-shrunk over rib skeleton, bonded to spars and trailing edge with contact adhesive.
- Trailing edge: 0.5 mm carbon strip, full span, bonded to rib trailing edges.
- VG array: bonded to upper membrane surface at correct chord position before membrane is tensioned.
- Covert flap rows: hinged to rib upper faces using piano wire, calibrated spring preload applied and tested before membrane installation.
- End plate: bonded and through-bolted to tip rib.
- Wing fences: bonded to upper membrane surface across rib-to-rib span.
- Root attachment plate: 2 mm carbon bonded and bolted to root rib and LE spar — provides interface to root mechanism gimbal.

CHAPTER 7

ELECTRONICS AND CONTROL SYSTEM

7.1 System Architecture

The electronic control system is a series architecture. The flight computer runs the kinematic wave generator and attitude stabilisation algorithms, outputting 24 independent servo position commands at 50 Hz update rate. These commands are distributed to the servo hardware through two daisy-chained PWM driver boards, each controlling 12 servo channels. The IMU provides attitude feedback at 200 Hz to the stabilisation loops.

7.2 Flight Computer

Unit	Raspberry Pi 5 (4 GB RAM variant)
Operating system	Raspberry Pi OS Lite 64-bit
Primary language	Python 3.11 with NumPy for waveform generation
Update rate	50 Hz main loop (servo output); 200 Hz IMU sampling
Weight	Approximately 70 g including heatsink
Power consumption	5V, 3–5 A typical under flight load
Justification	Sufficient processing power for 24-channel kinematic computation at 50 Hz; large community support; I2C and GPIO native

7.3 Servo Driver Boards

Unit	Adafruit PCA9685 16-Channel 12-bit PWM Driver (x2)
Interface	I2C bus — chained on same bus with different I2C addresses
Channels used	12 channels per board (24 total for 8 wings x 3 axes)

Resolution	12-bit (4096 steps) per channel — approximately 0.5 degree resolution
Update frequency	50 Hz (period 20 ms) — standard servo PWM
Output voltage	Servo power rail at 7.4V from dedicated BEC
Address configuration	Board 1: 0x40 (default); Board 2: 0x41 (A0 pin bridged)
Weight (each)	Approximately 9 g
Library	adafruit-circuitpython-servokit — Python, well maintained

7.4 Inertial Measurement Unit

Unit	Bosch BNO085 (Adafruit breakout board)
Interface	I2C at 400 kHz
Output	Fused attitude quaternion at up to 1000 Hz
Operating rate	200 Hz in this application
Accuracy	Heading: 2 degrees RMS; roll/pitch: 1 degree RMS
Weight	Approximately 3 g
Mounting	Central fuselage spine, vibration-isolated on 3 mm foam tape
Justification	Integrated sensor fusion — no external fusion algorithm required; calibrated factory — immediately usable

7.5 RC Receiver and FPV

RC receiver	ExpressLRS 2.4 GHz receiver (e.g. BetaFPV ELRS Lite)
Protocol	CRSF (Crossfire) serial — single wire to Pi GPIO UART
Channels used	5: Roll, Pitch, Throttle, Yaw, Wave Mode selector

FPV camera	RunCam Thumb Pro or similar — 4K, 16 g
FPV transmitter	Caddx Vista or DJI O3 Air Unit (optional — adds 35 g)
Telemetry	Pi transmits battery voltage, IMU attitude, wing fault status via CRSF telemetry back to transmitter

7.6 Wiring Architecture

- Main power bus: 22 AWG silicone wire, 6S LiPo positive to main power distribution board
- BEC output (7.4V): powers all 24 servos via PCA9685 servo power rail
- BEC output (5V): powers Raspberry Pi 5 via USB-C power delivery
- I2C bus: 3.3V logic — Pi GPIO2/GPIO3 to BNO085, PCA9685 board 1, PCA9685 board 2
- UART: Pi GPIO14/GPIO15 to ELRS receiver CRSF input
- All signal wires twisted pairs or shielded where routing near power cables
- XT60 connector on main battery lead; XT30 on BEC outputs
- Total wiring mass estimate: 45–65 g

CHAPTER 8

FIRMWARE ARCHITECTURE

8.1 Software Layers

The flight firmware is structured as four independent software layers, each running at an appropriate update rate:

Layer	Rate	Function
Layer 1: RC Parser	50 Hz	Reads CRSF serial, normalises 5 channels to -1.0 to +1.0 range, applies deadband
Layer 2: IMU Reader	200 Hz	Reads BNO085 quaternion, converts to Euler angles, detects faults
Layer 3: PID Stabiliser	100 Hz	Three independent PID loops: roll, pitch, yaw. Outputs attitude correction
Layer 4: Wave Generator	50 Hz	Computes per-wing phase offsets, sweep schedules, AoA profiles, outputs 24 servo angles

8.2 Wave Generator Detail

The wave generator is the core of the flight firmware. At each 50 Hz tick it performs the following computation for all 8 wings:

- Advance the master clock phase by (360 x flap_frequency / 50) degrees
- Compute each wing's cycle position: wing_phase = (master_phase + phase_offset[wing]) mod 360
- Retrieve flight mode parameters: base AoA, stroke amplitude, wave mode
- Apply RC mixing: roll (differential amplitude), pitch (fore/aft AoA), yaw (phase shift), throttle (amplitude scale)
- Apply sweep schedule: compute current sweep angle from cycle position and flight mode
- Compute kinematic output: stroke angle phi from sin(wing_phase), pitch angle psi from smoothed square wave profile, deviation angle theta from 2x frequency sinusoid
- Apply PID corrections to per-wing parameters

• Convert all angles to servo microsecond values and write to PCA9685 boards via I2C

8.3 Pitch Flip Profile

The pitch axis does not follow a sinusoidal profile. It uses a smoothed square wave — holding near maximum AoA during the power stroke and near zero AoA during recovery, with a rapid cubic-smoothstep transition at each stroke reversal. The flip zone is set to 20 degrees either side of stroke reversal, giving approximately 55 ms for the flip at 1 Hz. This is within the capability of the specified pitch servos (0.05 sec per 60 degrees).

8.4 Fault Detection and Compensation

The firmware monitors servo load feedback (where available) and IMU attitude divergence to detect wing failures. On detection of a failed wing, the firmware:

- Marks the failed wing as inactive — its servo outputs freeze at neutral
- Recalculates phase offsets for remaining active wings, redistributing evenly
- Increases stroke amplitude on the mirror wing on the opposite side to compensate
- Activates asymmetric AoA correction to counteract roll/yaw bias from asymmetric lift
- Alerts the pilot via telemetry — LED flash code and RSSI text warning
- The aircraft remains controllable with a minimum of 6 of 8 wings operational

CHAPTER 9

POWER SYSTEM

9.1 Main Battery

Chemistry	Lithium Polymer (LiPo)
Configuration	6S (6 cells in series) — 22.2V nominal
Capacity	5000–6000 mAh
Discharge rating	25C continuous minimum — supports peak servo demand
Dimensions (typical)	155 x 52 x 40 mm for 6000 mAh pack
Weight	560–680 g depending on capacity and manufacturer
Connector	XT60 male, 12 AWG silicone leads
Balance connector	XH 7-pin for cell-level charging and monitoring
Recommended brands	Tattu, Turnigy Graphene, CNHL — aviation-grade LiPo
Estimated hover endurance	18–32 minutes depending on flight mode and MTOW

9.2 Power Distribution

Main distribution board	XT60 input, multiple BEC outputs
Servo BEC (7.4V)	Minimum 10A continuous — powers all 24 servos
Logic BEC (5V)	3A — powers Raspberry Pi 5 and ancillary electronics
Voltage monitoring	Cell-level voltage divider to Pi GPIO analogue input or dedicated voltage monitor IC
Low voltage cutoff	Software warning at 3.6V per cell; land immediately below 3.5V per cell

Current monitoring	Inline current sensor on main bus — logged for power analysis

9.3 Power Budget

24 servos — hover average	~38–52 W (1.6–2.2 A per servo average at 7.4V)
24 servos — burst (takeoff)	~85–110 W peak, brief duration
Raspberry Pi 5	~8–12 W at 5V under flight computation load
IMU, receiver, FPV	~4–6 W combined
Total hover power	~50–70 W continuous
Total peak power	~100–130 W burst
Endurance calculation	74 Wh (6S 5000 mAh) divided by 60 W average = 74 minutes theoretical; actual approximately 30 minutes accounting for efficiency, reserves, and peak demands

CHAPTER 10

BUILD SEQUENCE AND MATERIALS

10.1 Recommended Build Order

The following sequence minimises rework and allows incremental testing at each stage:

Phase 1: Wing Construction

- CNC cut all rib profiles from 0.8 mm CFRP sheet — 8 wing sets
- Cut all leading edge spars (4 mm carbon tube) to span lengths
- Cut secondary spars (3 mm rod) to span lengths
- Assemble rib-and-spar skeleton for Wing 1 — trial fit and adjust
- Bond VG array to upper rib faces before membrane application
- Install covert flap piano wire hinges and calibrate spring preload
- Apply ripstop nylon membrane — heat shrink to tension
- Bond trailing edge carbon strip
- Bond end plates and wing fences
- Install root attachment plate
- Repeat for Wings 2, 3, 4 with respective geometry variations
- Build right-side mirrors of all four wings

Phase 2: Root Mechanisms

- CNC cut all gimbal frames from 3 mm CFRP plate
- Press-fit bearings into all pivot points
- Assemble sweep axis gimbals and calibrate sweep range stops
- Install flap axis gimbals — verify full 120 degree arc without binding
- Install pitch axis — verify rapid flip capability
- Install servos in each axis — bench test range of motion
- Fabricate and install push rods with ball link ends
- Attach wings to root mechanisms — verify no binding across full DoF range

Phase 3: Fuselage Assembly

- Cut carbon spine tube to length
- CNC cut root attachment frames — 8 frames per side

- Mark and drill root attachment positions on spine — verify stagger geometry
- Bond and bolt root attachment frames to spine
- Install cross-members connecting left and right sides
- Install equipment tray in electronics bay
- Install battery bay with access hatch
- Fit landing skids

Phase 4: Electronics Integration

- Install Raspberry Pi 5 on equipment tray with vibration isolation
- Mount PCA9685 boards adjacent to Pi — verify I2C addresses
- Install BNO085 IMU on spine with foam isolation
- Install power distribution board and BECs
- Route and connect all servo wiring — label all 24 channels
- Install RC receiver — bind to transmitter and verify signal
- Install FPV camera if fitted
- Connect battery and verify all voltages

Phase 5: Software and Calibration

- Install Raspberry Pi OS and required Python libraries
- Deploy flight firmware — verify basic servo movement on bench
- Calibrate servo centre positions and ranges for all 24 channels
- Calibrate IMU — run BNO085 calibration sequence
- Set up RC channel mapping and verify all 5 channels
- Run simulation mode — verify metachronal wave pattern visually
- Set PID initial gains conservatively
- Bench test fault detection — manually disconnect servos and verify response

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10.2 Materials Bill of Quantities

Material	Quantity	Application
Carbon fibre tube 4 mm OD	5 metres	Wing leading edge spars
Carbon fibre rod 3 mm	3 metres	Wing secondary spars, push rods
Carbon fibre sheet 0.8 mm	0.4 m sq	Wing ribs — all 8 wings

Material	Quantity	Application
Carbon fibre sheet 0.5 mm	0.2 m sq	End plates, wing fences, covert flaps
Carbon fibre sheet 2 mm	0.15 m sq	Root attachment plates, gimbal frames base
Carbon fibre sheet 3 mm	0.20 m sq	Gimbal frames, fuselage cross members
Carbon fibre tube 25x10 mm	0.75 metres	Fuselage spine
Ripstop nylon 20 micron	2.0 m sq	Wing membranes — all 8 wings
Piano wire 0.5 mm	3 metres	Covert flap hinges
Silver steel rod 3 mm	0.3 metres	Gimbal pivot pins
Ball bearings 3x7x3 mm	48 minimum	Gimbal pivot bearings
M2 M3 stainless hardware	Assorted	Fasteners throughout
Structural epoxy (Araldite 2011)	200 ml	CFRP bonding
Film adhesive (3M 9460PC)	1 m sq	VG and surface bonding

CHAPTER 11

TESTING AND COMMISSIONING

11.1 Bench Testing (Pre-Flight)

- Static load test: suspend aircraft inverted and apply downward load to wing tips — verify structural integrity to 2x estimated flight loads
- Full DoF range test: manually command each axis of each wing through full range — verify no binding, interference, or unexpected motion
- Wave pattern visual check: run metachronal wave at 0.5 Hz with aircraft suspended — verify phase relationships visually match specification
- Servo current monitoring: measure per-servo current at peak load — verify within BEC capacity
- IMU response test: move aircraft by hand, verify attitude readout is accurate and responsive
- Fault injection test: disable individual servo PWM channels, verify firmware detects and compensates
- Battery current draw: measure total current at bench with all servos active — verify against power budget

11.2 Tethered Hover Test

Before free flight, the aircraft should complete a tethered hover test. The aircraft is suspended on a vertical guide wire allowing vertical movement only, preventing lateral or forward flight while allowing genuine lift measurement.

- Confirm aircraft generates sufficient lift to suspend its own weight at mid-throttle
- Measure actual hover power draw and compare to estimate
- Verify attitude stabilisation loops hold level attitude with small perturbations
- Verify wave mode transitions are smooth and do not cause attitude disturbances
- Verify emergency glide mode — wings lock to fixed angle on throttle cut

11.3 Initial Free Flight Programme

Flight	Test Profile	Objective
Flight 1	Hover at 0.5 m, 60 seconds	Confirm basic stability, PID tuning

Flight	Test Profile	Objective
Flight 2	Hover to 2 m and back, 3 minutes	Confirm altitude control authority
Flight 3	Slow forward flight circuit	Confirm transition from hover to cruise
Flight 4	Wave mode switching in hover	Confirm mode transitions are smooth
Flight 5	Reverse flight demonstration	Confirm rearward translation
Flight 6	Wing stow test — reduce to 6 wings	Confirm partial stow flight capability
Flight 7	Maximum speed run	Characterise forward speed envelope
Flight 8+	Extended performance characterisation	Endurance, climb rate, gust testing

CHAPTER 12

PERFORMANCE TARGETS

12.1 Design-Stage Performance Estimates

The following performance parameters represent design-stage estimates derived from first-principles aerodynamic calculation, published research on analogous flapping wing systems, and extrapolation from validated data. Confidence ratings reflect the quality of the underlying evidence. All figures are subject to revision following physical test data acquisition.

PHYSICAL		
Maximum takeoff weight	2.8 – 4.2 kg	HIGH — scaled from component mass budget
Total wingspan	2.1 – 2.4 m	HIGH — from wing geometry specification
Total wing area (8 wings)	0.42 – 0.58 m sq	HIGH — from chord and span specification
Wing loading	58 – 73 N per m sq	HIGH — MTOW divided by wing area
AERODYNAMIC		
Peak lift coefficient (W1)	CL 2.4 – 2.6	HIGH — S1223 + VG + covert flap validated data
Cruise lift to drag	18 – 28	MEDIUM — extrapolated, unsteady effects estimated
Hover power loading	6 – 9 g per W	MEDIUM — cascade adds 15% over single-pair baseline
Hover figure of merit	0.62 – 0.74	MEDIUM — 120 degree stroke optimal amplitude
Effective AoA range	0 to 42 degrees	HIGH — VG + covert flap combination

FLIGHT		
Hover	Sustained, stable	HIGH — architecture proven at this mass range
Hover power draw	38 – 62 W	MEDIUM — from power loading and mass
Hover endurance	18 – 32 minutes	MEDIUM — 74 Wh battery at 60 W average
Minimum speed	0 km/h (VTOL)	HIGH — inherent in design
Cruise speed	18 – 28 km/h	MEDIUM — advance ratio analysis
Maximum speed	38 – 52 km/h	MEDIUM — swept back, glide mode
Glide ratio	8:1 to 12:1	MEDIUM — airfoil efficiency at Re range
Vertical climb rate	1.4 – 2.8 m/s	MEDIUM — excess power at MTOW
Reverse flight speed	0 – 8 km/h	MEDIUM — wave reversal, less efficient
Lateral translation	0 – 4 km/h	MEDIUM — asymmetric stroke plane tilt

CONTROL		
Roll rate maximum	35 – 65 degrees/s	MEDIUM — 3 mechanisms, 1 Hz bandwidth limit
Yaw rate maximum	45 – 80 degrees/s	MEDIUM — phase differential most effective axis
Gust rejection	Up to 6 m/s	MEDIUM — 8 wings, 3-mechanism redundancy
Fault tolerance	6 of 8 wings minimum	MEDIUM — phase recalculation verified in firmware

ACOUSTIC

Dominant aero frequency	1 Hz — inaudible	HIGH — below 20 Hz hearing threshold
Noise at 5 m	Less than 45 dB(A)	MEDIUM — no rotating disc or propeller

NOTE: All performance figures are design-stage estimates only. Actual performance will depend on build quality, component selection, manufacturing precision, aerodynamic surface finish, tuning of PID controllers, and environmental conditions. A systematic flight test programme is required to establish actual performance data. These figures represent first-iteration targets.

CHAPTER —

DOCUMENT NOTES

This engineering proposal is a design-stage document. It consolidates the aerodynamic analysis, component selection, build methodology, and performance estimates developed throughout the design process into a structured build specification. It does not constitute a certified design or approved airworthiness document. The aircraft described is an experimental unmanned aerial vehicle intended for research and development purposes.

The builder accepts full responsibility for compliance with local regulations governing unmanned aerial vehicles, including but not limited to registration, operational limitations, and insurance requirements. In most jurisdictions a UAV of this size and mass (2.8–4.2 kg) will require registration and may require operator certification. These requirements should be confirmed before first flight.

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